

TC ID	Module	Test Case Description	Expected Result	Result (Pass/Fail)
TC01	JSON File	Create new user in JSON file	User data stored successfully	
TC02	JSON File	Register with existing email	Duplicate registration prevented	
TC03	JSON File	JSON File Verification	JSON file does not exist handling	
TC04	JSON File	Empty JSON file	System handles properly	
TC05	JSON File	Corrupted JSON file	Error message displayed	
TC06	JSON File	Invalid JSON format	Validation error	
TC07	JSON File	Add 1000 users to JSON file	Data stored correctly	
TC08	JSON File	Read user data from JSON	Correct information displayed	
TC09	JSON File	Update user information	JSON updated successfully	
TC10	JSON File	Delete user information	Data removed correctly	
TC11	Email Scheduling	Schedule email	Stored in JSON	
TC12	Email Scheduling	Multiple scheduled emails	All stored correctly	
TC13	Email Scheduling	Edit scheduled email	JSON updated	
TC14	Email Scheduling	Delete scheduled email	Removed from JSON	
TC15	Email Scheduling	Restart application	Scheduled emails remain available	
TC16	Email Scheduling	Invalid date/time	Validation error	
TC17	Email Scheduling	500 scheduled emails	System handles successfully	
TC18	Email Sending	Send single email	Email delivered	
TC19	Email Sending	Send bulk emails	All emails delivered	
TC20	Email Sending	Invalid recipient email	Error shown	
TC21	Email Sending	Empty recipient field	Validation message	
TC22	Email Sending	Empty subject	Warning shown	
TC23	Email Sending	Empty body	Warning shown	
TC24	Email Sending	Internet disconnected	Error logged	
TC25	Email Sending	Gmail authentication failure	Proper error message	
TC26	AI Voice Reader	Read email aloud	Voice generated	
TC27	AI Voice Reader	Read long email	Entire email read	
TC28	AI Voice Reader	Read multilingual email	Correct pronunciation	
TC29	AI Voice Reader	Empty email content	Appropriate message	
TC30	AI Voice Reader	Voice engine unavailable	Error handled	
TC31	Spam Detection	Detect spam keywords	Mark as spam	
TC32	Spam Detection	Delete spam automatically	Spam removed	
TC33	Spam Detection	Legitimate email	Not deleted	
TC34	Spam Detection	Bulk spam emails	Processed correctly	
TC35	Spam Detection	Spam log file updated	Record maintained	
TC36	Graph & Report	Generate sent email graph	Graph displayed	
TC37	Graph & Report	Generate spam statistics graph	Accurate data shown	
TC38	Graph & Report	No records available	Proper message displayed	
TC39	Graph & Report	Export report	File generated	
TC40	Graph & Report	Large dataset graph	Generated successfully	
TC41	Security	Password stored in plain text	Should fail security check	
TC42	Security	Password hashing verification	Password stored securely	
TC43	Security	Unauthorized JSON file access	Access denied	
TC44	Security	Modify JSON manually	Validation performed	
TC45	Security	Multiple failed login attempts	Account temporarily blocked	
TC46	Performance	1000 users in JSON file	Fast retrieval	
TC47	Performance	1000 scheduled emails	Stable performance	
TC48	Performance	Run continuously for 24 hours	No crash	
TC49	Performance	Large JSON file (>10 MB)	Proper functionality	
TC50	Performance	Multiple users accessing data	Data integrity maintained	
TC51	Notification	Notification sent after successful email delivery	Notification displayed	
TC52	Notification	Notification sent for failed email	Failure alert shown	
TC53	Notification	Multiple notifications generated simultaneously	All notifications displayed	
TC54	Notification	Notification data stored in JSON	Stored correctly	
TC55	Notification	Clear notification history	Notifications removed	
TC56	Email Templates	Create new template	Saved in JSON	
TC57	Email Templates	Edit template	Updated successfully	
TC58	Email Templates	Delete template	Removed successfully	
TC59	Email Templates	Use template while sending email	Email generated correctly	
TC60	Email Templates	Template with missing placeholders	Validation warning	
TC61	Recipient Management	Add recipient	Stored in JSON	
TC62	Recipient Management	Duplicate recipient entry	Prevented	
TC63	Recipient Management	Delete recipient	Removed successfully	
TC64	Recipient Management	Import large recipient list	Imported correctly	
TC65	Recipient Management	Invalid email format in recipient list	Rejected	
TC66	Tracking	Track email open status	Status updated	
TC67	Tracking	Track email delivery status	Delivery recorded	
TC68	Tracking	Generate tracking report	Report displayed	
TC69	Tracking	Tracking data stored in JSON	Saved correctly	
TC70	Tracking	Tracking for bulk emails	All statuses updated	
TC71	JSON Data Integrity	Simultaneous JSON write operations	No data loss	
TC72	JSON Data Integrity	Sudden shutdown during JSON write	Recovery possible	
TC73	JSON Data Integrity	Missing JSON key	Default value handled	
TC74	JSON Data Integrity	Extra unexpected JSON field	Ignored safely	
TC75	JSON Data Integrity	Large JSON structure	Parsed correctly	
TC76	Edge Cases	User enters 500-character subject	Accepted	
TC77	Edge Cases	User enters 10,000-character body	Accepted	
TC78	Edge Cases	Subject contains emojis	Displayed correctly	
TC79	Edge Cases	Recipient name contains special characters	Accepted	
TC80	Edge Cases	Empty recipient list for bulk send	Validation error	
TC81	Backup & Recovery	Create JSON backup	Backup file generated	
TC82	Backup & Recovery	Restore backup	Data restored	
TC83	Backup & Recovery	Backup file missing	Proper error message	
TC84	Backup & Recovery	Corrupted backup file	Recovery fails safely	
TC85	Backup & Recovery	Automatic backup scheduling	Backup created on time	
TC86	AI Features	AI summarizes long email	Summary generated	
TC87	AI Features	AI identifies urgent emails	Marked as priority	
TC88	AI Features	AI categorizes emails	Correct category assigned	
TC89	AI Features	AI detects phishing content	Warning generated	
TC90	AI Features	AI voice reads summarized email	Audio generated	
TC91	Dashboard Analytics	Show total emails sent	Accurate count	
TC92	Dashboard Analytics	Show failed emails	Accurate count	
TC93	Dashboard Analytics	Show spam statistics	Correct values	
TC94	Dashboard Analytics	Refresh dashboard data	Updated instantly	
TC95	Dashboard Analytics	Dashboard with empty data	Proper message displayed	
TC96	Stress Testing	Store 10,000 recipients in JSON	No corruption	
TC97	Stress Testing	Schedule 5,000 emails	Stable operation	
TC98	Stress Testing	Send 1,000 emails in batch	Successful processing	
TC99	Stress Testing	Generate reports from large JSON files	Acceptable performance	
TC100	Stress Testing	Entire system running for 48 hours	No crashes or memory issues	